THE INCLUSION OF THE QUASI-COHERENT DERIVED CATEGORY HAS A RIGHT ADJOINT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A functor has a right adjoint precisely when every object sits in a distinguished triangle with a quasi-coherent first term and a last term orthogonal to all quasi-coherent objects. invoked at proof lines 5051-5080. Reduces the construction of the adjoint to producing such a triangle for every object.
- On an affine scheme the inclusion of the quasi-coherent derived category has the right adjoint given by the coherator. invoked at proof lines 5055-5062. Provides the base case of the induction.
- The induction principle on quasi-compact opens reduces the statement about a quasi-compact quasi-separated scheme to the base case and to a union of two quasi-compact opens. invoked at proof lines 5055-5063. Organises the proof into the base case and the union step.
- Pushforwards and pullbacks along open immersions preserve the quasi-coherent derived category, and the standard exact sequence relates the pushforwards from two opens and their intersection. invoked at proof lines 5081-5119. Produces the required triangle in the union step and the orthogonality computation.

Source reference. Original lemma 0CQS: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 5051-5060 for the reduction to the triangle criterion and again at proof lines 5090-5095 in the if and only if form; both uses are inside the counted span.

Source reference. Original lemma 0CQR: sources/stacks-chow/tags.txt, source lines 1–1. A navigation pointer in the adjoint construction; the construction carried out with it is inside the counted span.

Source reference. Original lemma 08DR: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5055-5060 to organise the proof into the base case and the union step; both are inside the counted span.

Source reference. Original remark 08D6: sources/stacks-perfect/pinned-original-source.tex, source lines 1543–1543. Cited at proof lines 5057-5060 for the existence

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of the right derived functor used in the base case; the base case argument is inside the counted span.

Source reference. Original lemma 08D9: sources/stacks-perfect/pinned-original-source.tex, source lines 1670–1670. Cited at proof lines 5061-5063 to settle the affine case; the induction set up that uses the base case is inside the counted span.

Source reference. Original lemma 08BV: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 5071-5080 for the displayed triangle in the union step; the verification of its properties is inside the counted span.

Source reference. Original lemma 08D5: sources/stacks-perfect/pinned-original-source.tex, source lines 661–661. Invoked at proof lines 5106-5110 to see that the pushed forward quasi-coherent term is quasi-coherent; the second required property is verified immediately after inside the counted span.

Source reference. Original lemma 08DW: sources/stacks-perfect/pinned-original-source.tex, source lines 581–581. Used at proof lines 5112-5118 to see that the pullback of a quasi-coherent object stays quasi-coherent, which combined with adjointness gives the vanishing; that computation is inside the counted span.

Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme. The inclusion functor $D_{QCoh}(\mathcal{O}_X) \to D(\mathcal{O}_X)$ has a right adjoint DQ_X.*

First proof. We will use the induction principle as in Cohomology of Schemes, Lemma 08DR to prove this. If $D(QCoh(\mathcal{O}_X)) \to D_{QCoh}(\mathcal{O}_X)$ is an equivalence, then the lemma is true because the functor RQ_X of Section 08D6 is a right adjoint to the functor $D(QCoh(\mathcal{O}_X)) \to D(\mathcal{O}_X)$. In particular, our lemma is true for affine schemes, see Lemma 08D9. Thus we see that it suffices to show: if $X = U \cup V$ is a union of two quasi-compact opens and the lemma holds for U, V, and $U \cap V$, then the lemma holds for X.

The adjoint exists if and only if for every object K of $D(\mathcal{O}_X)$ we can find a distinguished triangle

$$E' \to E \to K \to E'[1]$$

in $D(\mathcal{O}_X)$ such that E' is in $D_{QCoh}(\mathcal{O}_X)$ and such that $\mathrm{Hom}(M, K) = 0$ for all M in $D_{QCoh}(\mathcal{O}_X)$. See Derived Categories, Lemma 0CQS. Consider the distinguished triangle

$$E \to Rj_{U,*}E|_U \oplus Rj_{V,*}E|_V \to Rj_{U\cap V,*}E|_{U\cap V} \to E[1]$$

in $D(\mathcal{O}_X)$ of Cohomology, Lemma 08BV. By Derived Categories, Lemma 0CQR it suffices to construct the desired distinguished triangles for $Rj_{U,*}E|_U$, $Rj_{V,*}E|_V$, and $Rj_{U\cap V,*}E|_{U\cap V}$. This reduces us to the statement discussed in the next paragraph.

Let $j : U \to X$ be an open immersion corresponding with U a quasi-compact open for which the lemma is true. Let L be an object of $D(\mathcal{O}_U)$. Then there exists a distinguished triangle

$$E' \to Rj_*L \to K \to E'[1]$$

in $D(\mathcal{O}_X)$ such that E' is in $D_{QCoh}(\mathcal{O}_X)$ and such that $\mathrm{Hom}(M, K) = 0$ for all M in $D_{QCoh}(\mathcal{O}_X)$. To see this we choose a distinguished triangle

$$L' \to L \to Q \to L'[1]$$

in $D(\mathcal{O}_U)$ such that L' is in $D_{QCoh}(\mathcal{O}_U)$ and such that $\mathrm{Hom}(N, Q) = 0$ for all N in $D_{QCoh}(\mathcal{O}_U)$. This is possible because the statement in Derived Categories, Lemma 0CQS is an if and only if. We obtain a distinguished triangle

$$Rj_*L' \to Rj_*L \to Rj_*Q \to Rj_*L'[1]$$

in $D(\mathcal{O}_X)$. Observe that Rj_*L' is in $D_{QCoh}(\mathcal{O}_X)$ by Lemma 08D5. On the other hand, if M in $D_{QCoh}(\mathcal{O}_X)$, then

$$\mathrm{Hom}(M, Rj_*Q) = \mathrm{Hom}(Lj^*M, Q) = 0$$

because Lj^*M is in $D_{QCoh}(\mathcal{O}_U)$ by Lemma 08DW. This finishes the proof. □